PRANAY AGARWAL

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Analytical and detail-oriented Data Analyst with over 2 years of professional experience in data migration, SQL-based ETL pipelines, and dashboard reporting. Graduated from AI and Data Analytics programs at Durham College, building foundational and machine learning models tailored for banking applications. Proven track record of bridging the gap between technical data and non-technical stakeholders through clear, actionable visualizations in Tableau.

Skills

- **Programming Languages**: Python, SQL, R
- **Data Science & ML**: Machine Learning (Regression, Classification), Statistical Analysis, Feature Engineering, Hyperparameter Tuning, Explainable AI (SHAP), Data Cleaning & Processing
- **Data Visualization & BI**: Tableau, Excel
- **Cloud Platforms**: AWS, Salesforce (Sales Cloud, Service Cloud, Litify)
- **Tools & Databases**: Jupyter Notebook, Docker, APIs, MS SQL Server, Apache Spark, Hadoop, Git

Professional Experience

Data Analyst (Full-Time), Cloud Mentor Pvt Ltd **Dec 2020 - Jan 2023**

- Executed end-to-end data migration from legacy systems into Salesforce (Sales Cloud, Service Cloud, Litify) using Data Loader and Workbench, achieving 99.5% data integrity and reducing data loss by 90%
- Engineered automated ETL pipelines utilizing Python (Pandas, NumPy) and SQL to extract and transform data for AWS S3 and MS SQL Server databases, cutting manual processing effort by 30%
- Developed rigorous validation rules and testing plans in Python guarantee pre-go-live data accuracy, successfully decreasing post-migration support issues by 25%
- Designed highly customized dashboards and reports in Salesforce and Litify, accelerating reporting speed by 40% for cross-functional teams, including legal operations and business analysts
- Mentored 3 junior staff members on data quality best practices and collaborated with Salesforce admins, accelerating team onboarding and improving reporting accuracy by 15%

Data Analyst Intern (Full-Time), Bharti Airtel **May 2019 - Aug 2019**

- Analyzed massive customer behavior datasets using Python and SQL to identify key trends, delivering actionable insights that directly supported a 5% increase in customer retention
- Optimized big data workflows on Linux/Ubuntu environments by utilizing Apache Spark and Hadoop to process raw system logs, accelerating data preparation times by 20% for the broader analytics team
- Designed comprehensive data visualizations, charts, and dashboards to present analytical findings to key stakeholders, utilizing Git for version control and collaborative script management

Project

AI-Powered Medical Insurance Capstone Project, Durham College **Sept 2025 – Apr 2026**

- Developed and validated regression models that improved prediction accuracy for medical insurance claim costs by 15-25% (MAE/RMSE) over baseline models, leading to enhanced financial forecasting.
- Architected reusable regression workflows and tuned model settings with cross-validation to get more reliable, lower-error predictions.
- Used Explainable AI (SHAP) to identify key cost drivers, enabling more accurate risk assessment and policy optimization for insurers.
- Created an end-to-end data workflow—integrating SQL, Docker, APIs, and Tableau—improving data quality and cutting manual analysis time by 30–40%

Wireless Customer Churn Prediction, Durham College **Mar 2025 – Apr 2025**

- Developed a machine learning pipeline using Python (scikit-learn, pandas) to predict wireless customer churn, identifying key drivers like data plan usage and service call frequency.

- Recommended Logistic Regression for its interpretability, leading to a data-driven retention strategy that aims to reduce churn by 15% and increase lifetime customer value.

Highway Traffic Volume Prediction Capstone Project, Durham College **Jan 2025– Apr 2025**

- Engineered a machine learning model using Python and scikit-learn to forecast hourly traffic volume on westbound I-94, leveraging historical traffic, weather, and temporal data
- Evaluated and compared Linear Regression, Decision Tree, and Random Forest models, ultimately driving a 15% improvement in prediction accuracy over the baseline
- Designed and deployed an interactive Power BI dashboard to translate complex predictive model results into clear, actionable insights for non-technical stakeholders

Loan Approval Prediction, Durham College **Oct 2024 – Nov 2024**

- Achieved 84% accuracy and 0.91 ROC-AUC with a voting ensemble for loan approval prediction, demonstrating proficiency in data cleaning, EDA, feature engineering (e.g., debt-to-income ratio), class imbalance techniques (SMOTE), and model evaluation using Logistic Regression, Random Forest, and XGBoost.

Education

- **Graduate Certificate, Artificial Intelligence Analysis, Design & Implementation**
  Durham College, Oshawa, ON **2025 – 2026**
- Graduated with College Honour Roll.
- Coursework: AI Algorithm, Applied Machine Learning, Applied Math.
- **Graduate Certificate, Data Analytics for Business Decision Making** **2024 – 2025**
  Durham College, Oshawa, ON
- Graduated with President's Honour and College Honour Roll
- Coursework: Statistical and Predictive Analysis, Introduction to Data Analytics, Data Collection, Visualization.
- **Bachelor of Engineering, Electronics and Communication Engineering** **2016 - 2020**
  University of Delhi, Netaji Subhas Institute of Technology, New Delhi, India
  Coursework: Database Management Systems, Mathematical Statistics, Big Data and Analytics